

EffDim stability across UniverseTBD embedding samples

Experiment

Each dataset was evaluated four ways: the complete matrix; 100 row-bootstrap replicates; 25 uniform 10% samples without replacement; and 12 contiguous, nearly equal stored-order chunks. Repeated draws in each bootstrap replicate were merged into one unique row. All 15 non-GMST methods used streaming covariance and exact cuVS GPU neighbours ($k=10$).

The contiguous chunks test sensitivity to stored row order. They are not physical sky regions because the embedding matrices contain no coordinates.

Main findings

1. **The simplified ESS proxy has the lowest normalized spread, but it is not a calibrated dimension estimate.** Its median relative standard deviations are 0.116% for bootstrap samples, 0.512% for 10% samples, and 1.183% for regions. This indicates repeatability only.
2. **Large-dataset spectral estimates are stable.** On DESI and Legacy Survey, participation ratio and all Rényi dimensions remain close to the complete dataset under bootstrap, 10% sampling, and region splits. Region CVs are approximately 2–5%.
3. **JWST is too small for 12-way high-dimensional spectral comparisons.** Each region has only about 125 rows but 1,024 features, imposing a rank ceiling. PCA-95 falls from 145 to 47–55 and geometric-mean ED becomes numerically dominated by the null spectrum.
4. **Deduplicated bootstrap avoids artificial zero-distance clouds.** It behaves as a random unique subset containing about 63% of the rows, so remaining differences primarily reflect sample-size sensitivity.
5. **ESS and DANCo are the most region-stable geometry methods.** ESS region CV is 0.7–1.9% and DANCo is 1.2–3.4% across the three datasets.
6. **Agreement across regions does not imply agreement with the whole set.** MLE/TLE regions have low internal CV but systematically return only about 75–81% of the whole-dataset estimate, demonstrating sample-size bias.
7. **Two-NN is sensitive to ties already present in these embeddings.** Its Legacy Survey region CV is 53.5%, and the whole JWST estimate (0.115) is inconsistent with its region estimates (about 4–7).

Stability ranking

Lower relative standard deviation means better repeatability. The bias column is also required: a method can have low spread while being consistently far from its whole-dataset estimate. Values are medians across datasets, and the overall columns are medians across all three schemes and datasets.

Method	Bootstrap RSD	Sparse RSD	Region RSD	Overall RSD	Overall absolute bias
ESS	0.116%	0.512%	1.183%	0.512%	2.00%
PCA-95	0.351%	1.032%	2.155%	1.032%	5.45%
DANCo	0.226%	0.995%	1.907%	1.041%	5.92%
MLE	0.330%	0.748%	2.368%	1.306%	18.72%
TLE	0.330%	0.748%	2.368%	1.306%	18.72%
Shannon ED	0.357%	1.396%	3.455%	1.396%	3.01%
MiND-MLk	0.391%	0.999%	2.754%	1.485%	19.21%
Rényi $\alpha=2$	0.479%	1.848%	4.885%	1.848%	0.65%
Participation ratio	0.479%	1.848%	4.885%	1.848%	0.65%
Rényi $\alpha=3$	0.538%	2.039%	4.489%	2.039%	0.67%
Rényi $\alpha=5$	0.555%	2.060%	4.028%	2.060%	0.76%
Rényi $\alpha=4$	0.556%	2.081%	4.215%	2.081%	0.74%
Geometric mean ED	0.340%	1.599%	6.673%	3.980%	28.21%
Two-NN	4.050%	33.292%	33.521%	6.572%	18.46%
MiND-MLi	7.503%	12.156%	15.229%	12.038%	19.46%

Whole-dataset estimates

Method	JWST	DESI	Legacy Survey
PCA-95	145.000	64.000	110.000
Participation ratio	11.752	8.659	9.241
Shannon ED	36.134	22.157	25.981
Rényi $\alpha=2$	11.752	8.659	9.241
Rényi $\alpha=3$	8.334	6.405	6.978
Rényi $\alpha=4$	7.053	5.505	6.105
Rényi $\alpha=5$	6.388	5.027	5.633
Geometric mean ED	29.822	57.334	23.050
MLE	12.560	18.172	22.560
Two-NN	0.115	16.782	22.502
DANCo	4.533	5.591	5.620
MiND-MLi	0.201	0.789	0.747
MiND-MLk	11.464	16.539	20.712
ESS	0.211	0.224	0.222
TLE	12.560	18.172	22.560

Sampled dimensionality distributions

Each panel is a histogram of the raw reported dimensionalities. Short rug marks at the baseline show every individual observation, and the dashed vertical line is the whole-dataset estimate. Horizontal axes are logarithmic. The tables separately provide compact mean $\pm$ SD summaries.

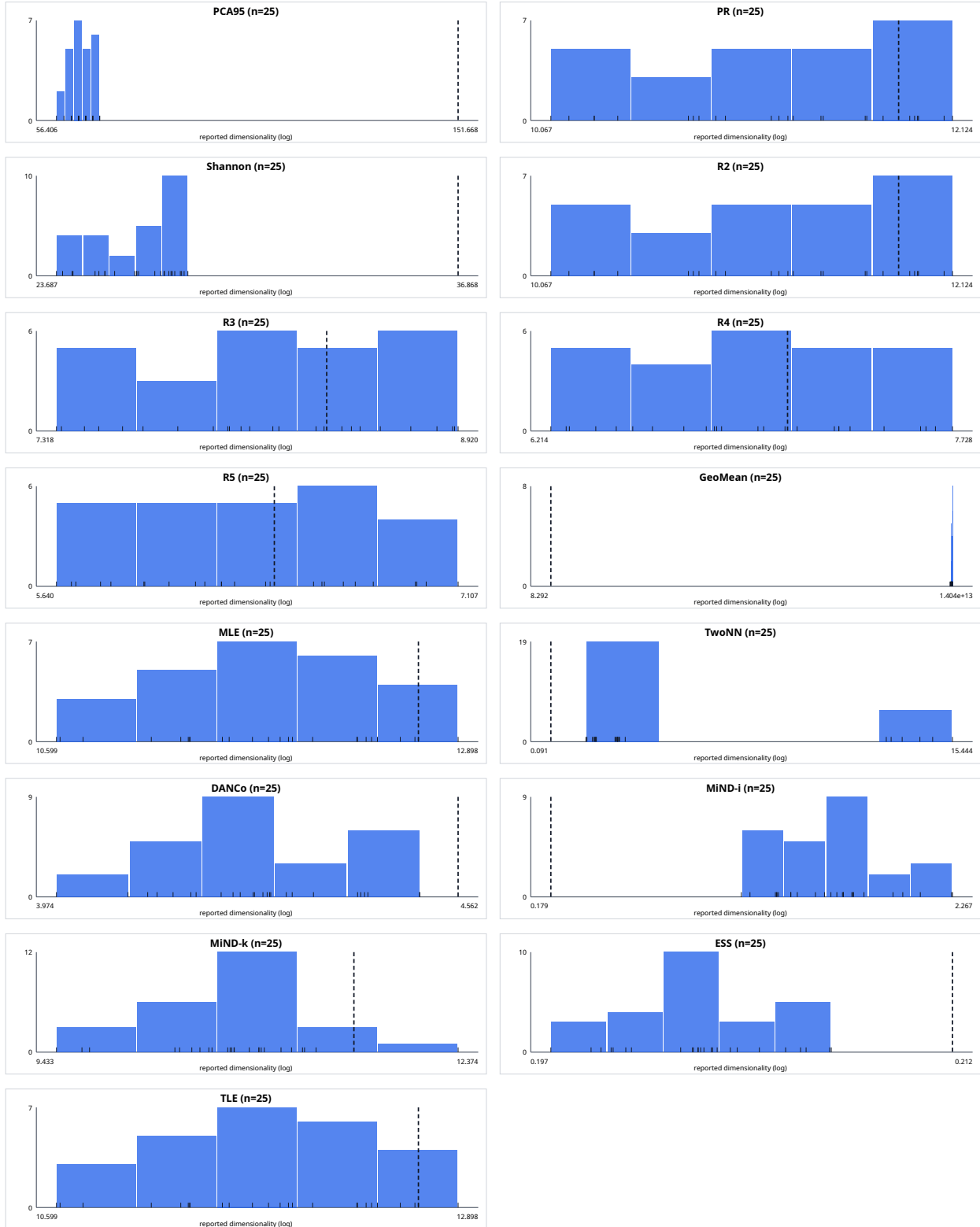
JWST: Bootstrap, duplicates merged

Histogram and rug marks show every raw observation; dashed line: whole-dataset estimate; logarithmic dimensionality axis



JWST: Sparse 10%

Histogram and rug marks show every raw observation; dashed line: whole-dataset estimate; logarithmic dimensionality axis



JWST: 12 regions

Histogram and rug marks show every raw observation; dashed line: whole-dataset estimate; logarithmic dimensionality axis



Method	Whole dataset	Bootstrap mean $\pm$ SD	Sparse mean $\pm$ SD	Regions mean $\pm$ SD
PCA-95	145.000	131.750 $\pm$ 1.344	62.360 $\pm$ 1.497	52.500 $\pm$ 2.4
Participation ratio	11.752	11.705 $\pm$ 0.176	11.175 $\pm$ 0.587	10.637 $\pm$ 0.6
Shannon ED	36.134	35.075 $\pm$ 0.467	26.188 $\pm$ 1.085	23.590 $\pm$ 1.2
Rényi $\alpha=2$	11.752	11.705 $\pm$ 0.176	11.175 $\pm$ 0.587	10.637 $\pm$ 0.6
Rényi $\alpha=3$	8.334	8.315 $\pm$ 0.133	8.137 $\pm$ 0.466	7.863 $\pm$ 0.5

Method	Whole dataset	Bootstrap mean $\pm$ SD	Sparse mean $\pm$ SD	Regions mean $\pm$ SD
Rényi $\alpha=4$	7.053	7.041 $\pm$ 0.119	6.944 $\pm$ 0.425	6.743 $\pm$ 0.425
Rényi $\alpha=5$	6.388	6.379 $\pm$ 0.111	6.318 $\pm$ 0.404	6.145 $\pm$ 0.388
Geometric mean ED	29.822	1.096e+03 $\pm$ 421.538	3.658e+12 $\pm$ 1.537e+11	9.762e+12 $\pm$ 3.089e+11
MLE	12.560	13.305 $\pm$ 0.272	11.777 $\pm$ 0.595	9.927 $\pm$ 0.812
Two-NN	0.115	0.134 $\pm$ 0.005	2.119 $\pm$ 3.634	5.176 $\pm$ 0.812
DANCo	4.533	4.505 $\pm$ 0.047	4.263 $\pm$ 0.121	4.276 $\pm$ 0.115
MiND-MLi	0.201	0.294 $\pm$ 0.024	1.082 $\pm$ 0.389	1.531 $\pm$ 0.201
MiND-MLk	11.464	12.054 $\pm$ 0.294	10.650 $\pm$ 0.538	8.702 $\pm$ 0.927
ESS	0.211	0.209 $\pm$ 0.001	0.202 $\pm$ 0.003	0.205 $\pm$ 0.001
TLE	12.560	13.305 $\pm$ 0.272	11.777 $\pm$ 0.595	9.927 $\pm$ 0.812

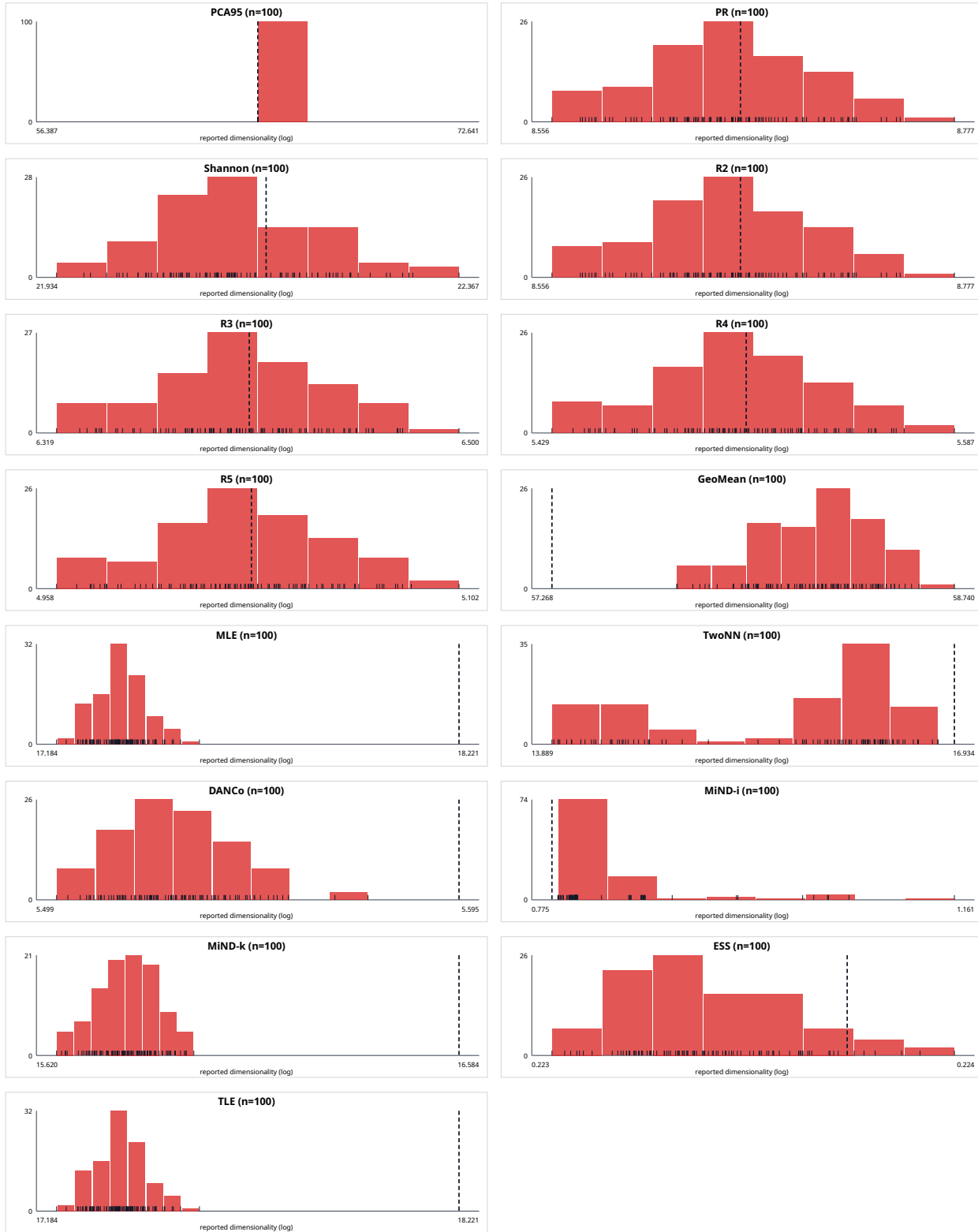
Individual contiguous-region estimates

Method	Whole	R1	R2	R3	R4	R5	R6
PCA-95	145.000	52.000	54.000	55.000	47.000	50.000	52.000
Participation ratio	11.752	10.698	10.778	10.242	9.915	10.175	12.264
Shannon ED	36.134	23.091	23.819	23.647	21.126	22.458	25.338
Rényi $\alpha=2$	11.752	10.698	10.778	10.242	9.915	10.175	12.264
Rényi $\alpha=3$	8.334	8.065	7.955	7.512	7.509	7.562	9.153
Rényi $\alpha=4$	7.053	6.984	6.787	6.442	6.500	6.503	7.815
Rényi $\alpha=5$	6.388	6.392	6.157	5.878	5.945	5.933	7.080
Geometric mean ED	29.822	9.390e+12	9.579e+12	9.766e+12	9.461e+12	9.561e+12	9.928e+12
MLE	12.560	9.706	10.847	10.182	8.738	8.922	8.956
Two-NN	0.115	5.675	6.100	4.972	4.557	4.462	4.885
DANCo	4.533	4.463	4.417	4.457	4.247	4.269	4.275
MiND-MLi	0.201	1.310	1.748	2.054	1.207	1.379	1.376
MiND-MLk	11.464	8.504	9.982	9.362	6.852	7.405	8.262
ESS	0.211	0.211	0.206	0.206	0.208	0.208	0.207
TLE	12.560	9.706	10.847	10.182	8.738	8.922	8.956

Method	R7	R8	R9	R10	R11	R12
PCA-95	53.000	53.000	55.000	50.000	54.000	55.000
Participation ratio	10.849	10.711	10.859	9.504	10.905	10.748
Shannon ED	24.178	24.118	24.739	21.837	24.036	24.693
Rényi $\alpha=2$	10.849	10.711	10.859	9.504	10.905	10.748
Rényi $\alpha=3$	7.996	7.856	7.883	6.998	8.121	7.741
Rényi $\alpha=4$	6.849	6.734	6.705	6.018	7.017	6.565
Rényi $\alpha=5$	6.234	6.145	6.086	5.502	6.433	5.955
Geometric mean ED	9.445e+12	9.672e+12	9.760e+12	1.035e+13	1.004e+13	1.020e+13
MLE	10.225	10.517	10.467	8.939	10.827	10.801
Two-NN	5.672	4.006	5.126	4.176	6.556	5.923
DANCo	4.029	4.263	4.363	4.216	4.299	4.010
MiND-MLi	1.467	1.413	1.738	1.347	1.759	1.578
MiND-MLk	8.519	9.293	8.807	8.027	9.587	9.822
ESS	0.198	0.203	0.204	0.205	0.204	0.198
TLE	10.225	10.517	10.467	8.939	10.827	10.801

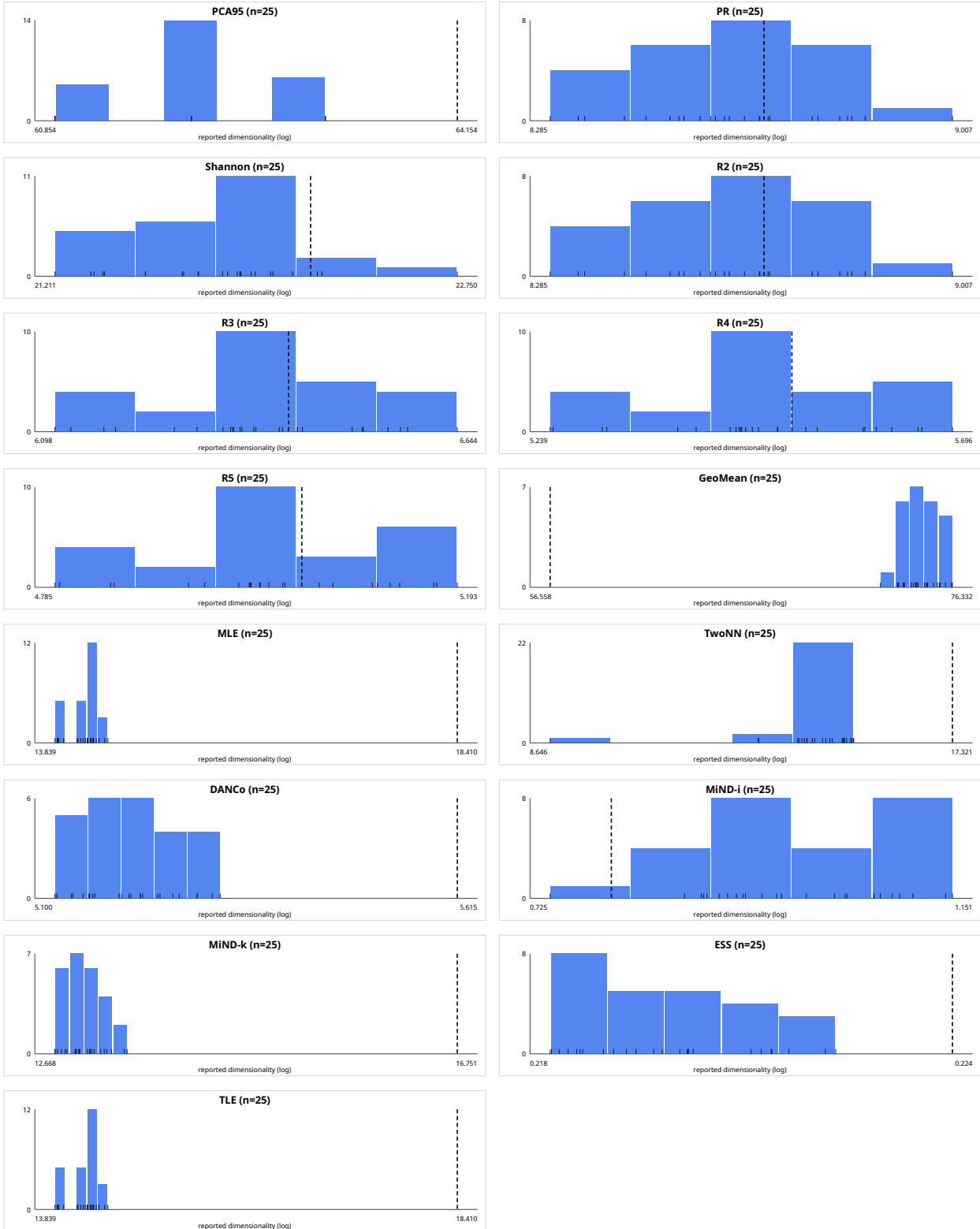
DESI: Bootstrap, duplicates merged

Histogram and rug marks show every raw observation; dashed line: whole-dataset estimate; logarithmic dimensionality axis



DESI: Sparse 10%

Histogram and rug marks show every raw observation; dashed line: whole-dataset estimate; logarithmic dimensionality axis



DESI: 12 regions

Histogram and rug marks show every raw observation; dashed line: whole-dataset estimate; logarithmic dimensionality axis



Method	Whole dataset	Bootstrap mean $\pm$ SD	Sparse mean $\pm$ SD	Regions mean $\pm$ SD
PCA-95	64.000	64.000 $\pm$ 0	62.040 $\pm$ 0.676	60.083 $\pm$ 1.379
Participation ratio	8.659	8.655 $\pm$ 0.042	8.614 $\pm$ 0.160	8.545 $\pm$ 0.229
Shannon ED	22.157	22.131 $\pm$ 0.079	21.823 $\pm$ 0.309	21.425 $\pm$ 0.542
Rényi $\alpha=2$	8.659	8.655 $\pm$ 0.042	8.614 $\pm$ 0.160	8.545 $\pm$ 0.229
Rényi $\alpha=3$	6.405	6.402 $\pm$ 0.034	6.372 $\pm$ 0.131	6.350 $\pm$ 0.186

Method	Whole dataset	Bootstrap mean $\pm$ SD	Sparse mean $\pm$ SD	Regions mean $\pm$ SD
Rényi $\alpha=4$	5.505	5.503 $\pm$ 0.031	5.477 $\pm$ 0.115	5.471 $\pm$ 0.172
Rényi $\alpha=5$	5.027	5.026 $\pm$ 0.028	5.002 $\pm$ 0.104	5.003 $\pm$ 0.162
Geometric mean ED	57.334	58.208 $\pm$ 0.195	73.659 $\pm$ 0.917	80.919 $\pm$ 2.282
MLE	18.172	17.376 $\pm$ 0.060	14.284 $\pm$ 0.136	13.705 $\pm$ 0.430
Two-NN	16.782	15.526 $\pm$ 0.859	13.483 $\pm$ 1.103	12.915 $\pm$ 0.516
DANCo	5.591	5.529 $\pm$ 0.013	5.205 $\pm$ 0.056	5.116 $\pm$ 0.107
MiND-MLi	0.789	0.830 $\pm$ 0.059	0.962 $\pm$ 0.096	1.010 $\pm$ 0.120
MiND-MLk	16.539	15.814 $\pm$ 0.065	13.075 $\pm$ 0.165	12.459 $\pm$ 0.456
ESS	0.224	0.223 $\pm$ 2.605e-04	0.220 $\pm$ 0.001	0.219 $\pm$ 0.003
TLE	18.172	17.376 $\pm$ 0.060	14.284 $\pm$ 0.136	13.705 $\pm$ 0.430

Individual contiguous-region estimates

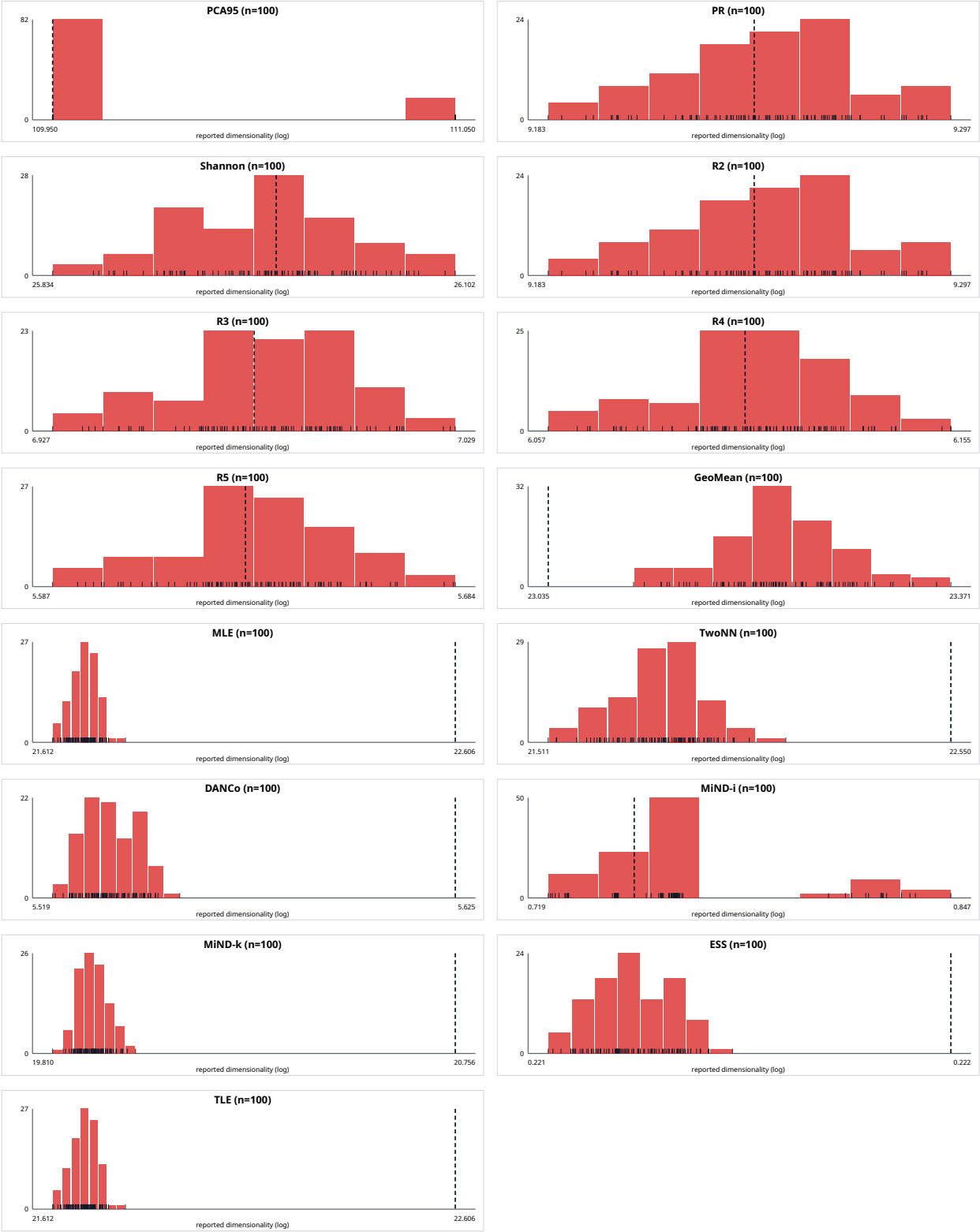
Method	Whole	R1	R2	R3	R4	R5	R6
PCA-95	64.000	61.000	60.000	60.000	61.000	61.000	63.000
Participation ratio	8.659	8.845	8.369	8.216	8.541	8.598	8.864
Shannon ED	22.157	22.182	21.133	21.072	21.602	21.894	22.345
Rényi $\alpha=2$	8.659	8.845	8.369	8.216	8.541	8.598	8.864
Rényi $\alpha=3$	6.405	6.521	6.191	6.054	6.312	6.314	6.536
Rényi $\alpha=4$	5.505	5.590	5.321	5.201	5.423	5.408	5.603
Rényi $\alpha=5$	5.027	5.098	4.862	4.753	4.952	4.932	5.108
Geometric mean ED	57.334	79.373	82.993	80.960	80.614	79.155	76.190
MLE	18.172	13.119	13.401	14.067	13.933	13.432	13.456
Two-NN	16.782	12.768	12.594	13.706	12.107	12.667	12.646
DANCo	5.591	5.216	5.121	5.177	5.045	5.175	5.270
MiND-MLi	0.789	0.955	0.875	1.038	0.882	1.106	1.155
MiND-MLk	16.539	11.887	12.106	12.832	12.752	12.234	11.992
ESS	0.224	0.222	0.220	0.219	0.218	0.221	0.223
TLE	18.172	13.119	13.401	14.067	13.933	13.432	13.456

Method	R7	R8	R9	R10	R11	R12
PCA-95	58.000	58.000	60.000	59.000	60.000	60.000
Participation ratio	8.519	8.472	8.471	8.258	8.925	8.466
Shannon ED	21.119	20.705	21.226	20.724	21.847	21.250
Rényi $\alpha=2$	8.519	8.472	8.471	8.258	8.925	8.466
Rényi $\alpha=3$	6.331	6.390	6.336	6.156	6.751	6.300
Rényi $\alpha=4$	5.452	5.547	5.479	5.318	5.872	5.435
Rényi $\alpha=5$	4.984	5.090	5.018	4.872	5.392	4.973
Geometric mean ED	83.335	83.617	79.750	84.043	80.033	80.968
MLE	13.267	13.655	14.321	14.021	14.414	13.381
Two-NN	13.298	13.026	12.174	13.335	13.041	13.613
DANCo	5.254	5.055	5.128	4.963	5.026	4.959
MiND-MLi	0.982	0.845	1.173	0.891	1.139	1.082
MiND-MLk	12.018	12.255	13.060	12.878	13.193	12.296
ESS	0.223	0.219	0.218	0.215	0.216	0.216
TLE	13.267	13.655	14.321	14.021	14.414	13.381

Legacy Survey

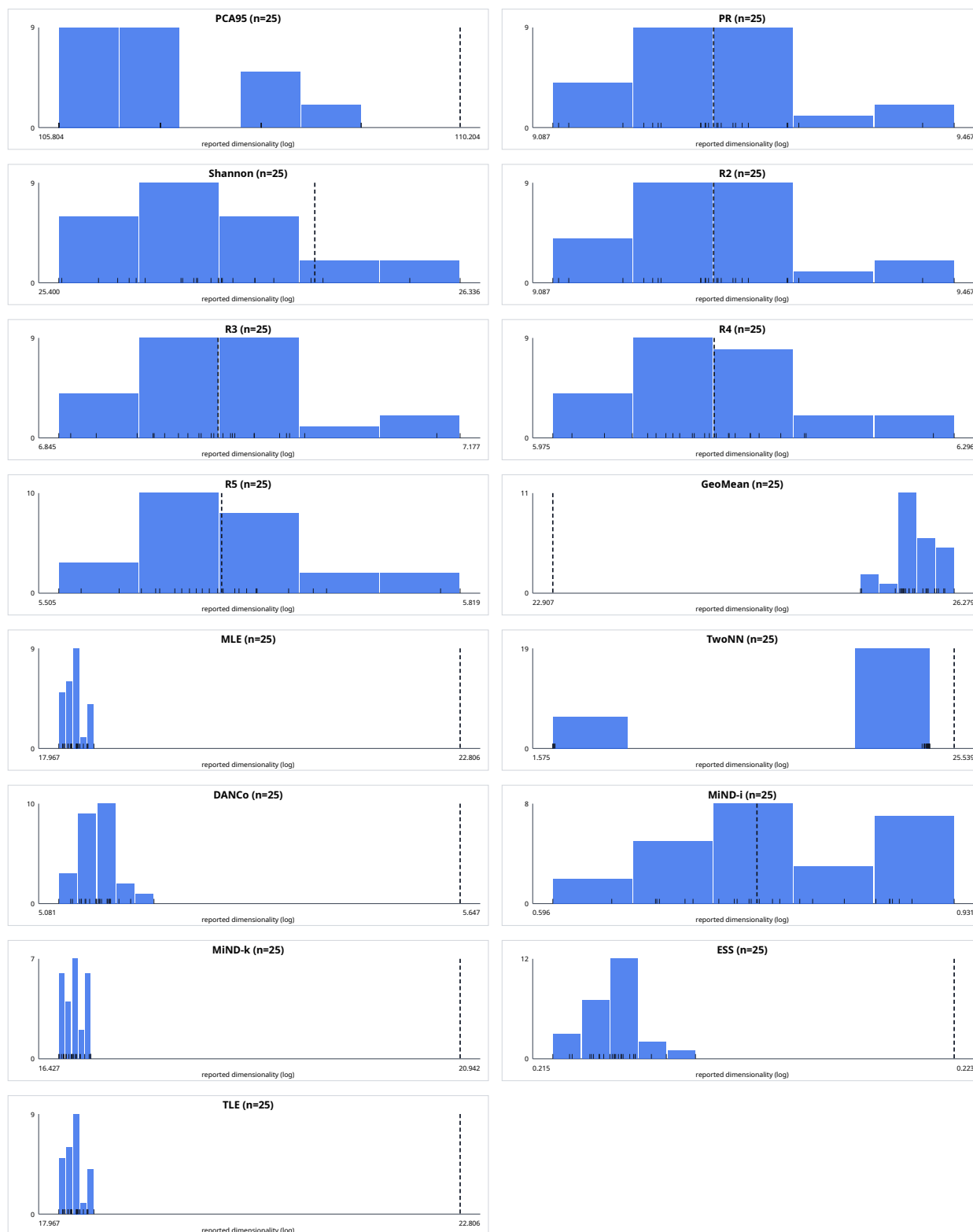
Legacy Survey: Bootstrap, duplicates merged

Histogram and rug marks show every raw observation; dashed line: whole-dataset estimate; logarithmic dimensionality axis



Legacy Survey: Sparse 10%

Histogram and rug marks show every raw observation; dashed line: whole-dataset estimate; logarithmic dimensionality axis



Legacy Survey: 12 regions

Histogram and rug marks show every raw observation; dashed line: whole-dataset estimate; logarithmic dimensionality axis



Method	Whole dataset	Bootstrap mean $\pm$ SD	Sparse mean $\pm$ SD	Regions mean $\pm$ SD
PCA-95	110.000	110.180 $\pm$ 0.386	107.000 $\pm$ 0.957	102.583 $\pm$ 5.435
Participation ratio	9.241	9.244 $\pm$ 0.023	9.240 $\pm$ 0.082	9.055 $\pm$ 0.451
Shannon ED	25.981	25.974 $\pm$ 0.052	25.765 $\pm$ 0.202	24.943 $\pm$ 1.355
Rényi $\alpha=2$	9.241	9.244 $\pm$ 0.023	9.240 $\pm$ 0.082	9.055 $\pm$ 0.451
Rényi $\alpha=3$	6.978	6.981 $\pm$ 0.020	6.980 $\pm$ 0.071	6.872 $\pm$ 0.313

Method	Whole dataset	Bootstrap mean $\pm$ SD	Sparse mean $\pm$ SD	Regions mean $\pm$ SD
Rényi $\alpha=4$	6.105	6.107 $\pm$ 0.019	6.107 $\pm$ 0.068	6.024 $\pm$ 0.257
Rényi $\alpha=5$	5.633	5.636 $\pm$ 0.018	5.636 $\pm$ 0.067	5.565 $\pm$ 0.227
Geometric mean ED	23.050	23.226 $\pm$ 0.048	25.799 $\pm$ 0.186	27.366 $\pm$ 1.538
MLE	22.560	21.726 $\pm$ 0.029	18.322 $\pm$ 0.087	17.730 $\pm$ 0.295
Two-NN	22.502	21.816 $\pm$ 0.101	14.854 $\pm$ 7.491	14.100 $\pm$ 7.543
DANCo	5.620	5.537 $\pm$ 0.006	5.154 $\pm$ 0.025	5.214 $\pm$ 0.062
MiND-MLi	0.747	0.760 $\pm$ 0.027	0.763 $\pm$ 0.080	0.805 $\pm$ 0.057
MiND-MLk	20.712	19.937 $\pm$ 0.032	16.752 $\pm$ 0.097	16.217 $\pm$ 0.308
ESS	0.222	0.222 $\pm$ 1.176e-04	0.217 $\pm$ 4.953e-04	0.218 $\pm$ 0.001
TLE	22.560	21.726 $\pm$ 0.029	18.322 $\pm$ 0.087	17.730 $\pm$ 0.295

Individual contiguous-region estimates

Method	Whole	R1	R2	R3	R4	R5	R6
PCA-95	110.000	99.000	104.000	107.000	111.000	102.000	96.000
Participation ratio	9.241	8.843	9.196	9.057	9.244	8.991	8.526
Shannon ED	25.981	24.291	25.352	25.583	26.174	24.789	23.203
Rényi $\alpha=2$	9.241	8.843	9.196	9.057	9.244	8.991	8.526
Rényi $\alpha=3$	6.978	6.714	6.972	6.792	6.925	6.849	6.519
Rényi $\alpha=4$	6.105	5.884	6.109	5.916	6.027	6.029	5.741
Rényi $\alpha=5$	5.633	5.432	5.641	5.444	5.543	5.588	5.320
Geometric mean ED	23.050	28.171	26.930	26.147	25.365	27.612	29.358
MLE	22.560	17.895	17.698	17.431	17.350	17.885	17.866
Two-NN	22.502	1.609	1.638	17.967	17.906	1.601	17.908
DANCo	5.620	5.172	5.234	5.227	5.138	5.163	5.214
MiND-MLi	0.747	0.836	0.738	0.865	0.761	0.748	0.762
MiND-MLk	20.712	16.378	16.166	15.925	15.786	16.321	16.419
ESS	0.222	0.217	0.219	0.219	0.218	0.217	0.218
TLE	22.560	17.895	17.698	17.431	17.350	17.885	17.866

Method	R7	R8	R9	R10	R11	R12
PCA-95	94.000	95.000	106.000	104.000	106.000	107.000
Participation ratio	8.415	8.527	9.482	8.916	9.577	9.889
Shannon ED	22.836	23.026	25.979	24.915	26.404	26.762
Rényi $\alpha=2$	8.415	8.527	9.482	8.916	9.577	9.889
Rényi $\alpha=3$	6.451	6.560	7.207	6.765	7.202	7.506
Rényi $\alpha=4$	5.692	5.802	6.323	5.935	6.274	6.563
Rényi $\alpha=5$	5.281	5.392	5.841	5.484	5.773	6.045
Geometric mean ED	29.798	29.684	26.421	26.572	26.190	26.152
MLE	17.982	18.070	17.930	18.020	17.374	17.260
Two-NN	19.362	18.743	18.341	18.379	17.490	18.258
DANCo	5.223	5.145	5.259	5.158	5.310	5.328
MiND-MLi	0.835	0.738	0.799	0.865	0.807	0.908
MiND-MLk	16.459	16.710	16.326	16.454	15.926	15.733
ESS	0.218	0.216	0.219	0.217	0.221	0.221
TLE	17.982	18.070	17.930	18.020	17.374	17.260